AirPods Pro 3

A research note on one pair of earbuds. Listening modes, the decibel figures Apple actually prints, a laboratory frequency curve, every stem gesture, what each microphone is for, and which speech features turn sound into text.

Announced 9 September 2025. In stores 19 September 2025. 249 US dollars. Apple H2 in each earbud, Bluetooth 5.3, IP57 on the earbuds and the case. Each earbud is 30.9 × 19.2 × 27.0 mm and weighs 5.55 g. Five foam-infused tips, XXS through L. Nothing in this note was measured on hardware in this project. Where Apple does not publish a number, the number is left out.

How to read the figures

Blue is Active Noise Cancellation. Teal is Transparency. Amber is Adaptive Audio, and also the dashed passive rating where that rating is the reference. Stone is the foam seal with the electronics off. The same colour means the same thing on every figure.

Three decibel tests are kept on separate figures. Apple's hearing-protection total is one number for a whole noisy room. Apple's octave table is the unpowered seal at one frequency. The SoundGuys curves are a single fixture sweep. A doubling of acoustic power is 3 dB. A sound that seems about half as loud is closer to 10 dB. Apple's "up to 2× more than AirPods Pro 2" is the power kind of claim, from an IEC 60268-24 test, and Apple does not publish that curve.

What active cancellation adds, by frequency

Blue fill is the gap between the seal alone and ANC on. Where the stone line rises above the blue, the foam is doing more than the electronics.

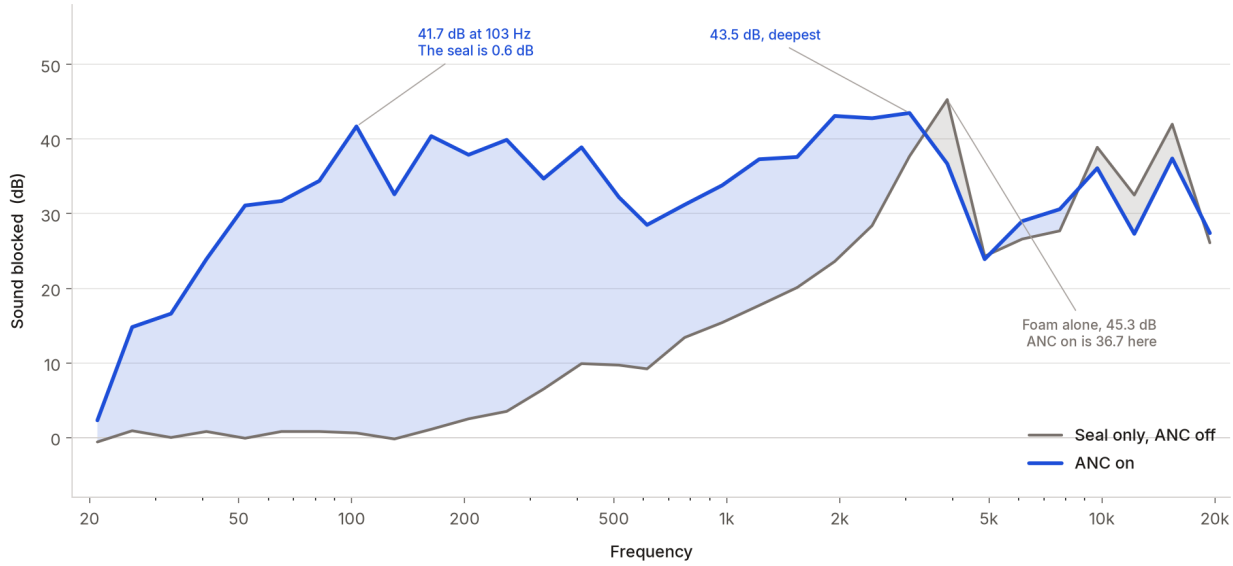


Figure 1 SoundGuys laboratory sweep on a B&K 5128, retrieved 5 November 2026. Not an Apple specification. One published sample.

Active cancellation is a low-frequency tool. Under about 200 Hz the foam blocks almost nothing, so the blue fill is nearly the whole of the protection. The deepest point on this sweep is 43.5 dB at 3.1 kHz. From about 4 kHz up, the seal is already strong, and at 3.9 kHz the foam alone measured 45.3 dB while ANC on measured 36.7 dB. Cancellation is not more blocking at every frequency.

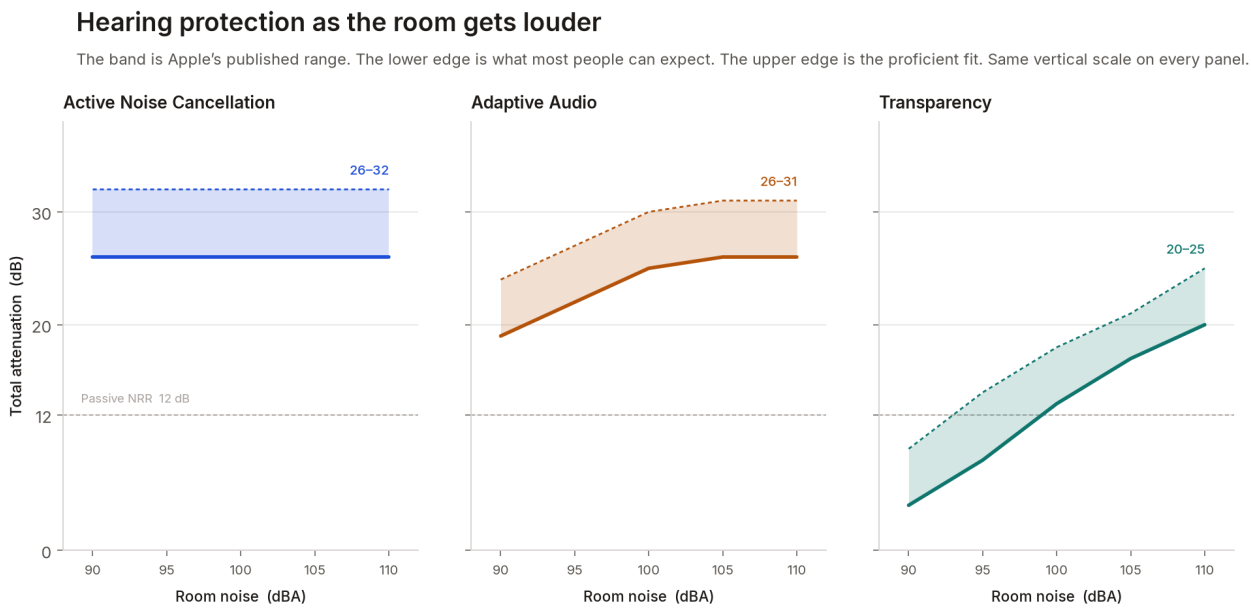


Figure 2 Apple Hearing Protection data sheet, AirPods Pro 3, September 2025. ANSI/ASA S12.68. Sustained noise only. Not for impulse sound, and not above 110 dBA.

ANC's estimated protection stays flat: 26 to 32 dB from a 90 dBA room through a 110 dBA room. Transparency starts near a light reduction, 4 to 9 dB at 90 dBA, and climbs to 20 to 25 dB at 110 dBA, because Loud Sound Reduction turns protection up as the room gets louder. Adaptive Audio sits between them and meets ANC once the room is very loud. The 12 dB dashed line is all that remains when the mode is Off or the battery is empty. These totals are for sustained noise. They are not for gunfire, fireworks, or jackhammers, and not for sound louder than 110 dBA.

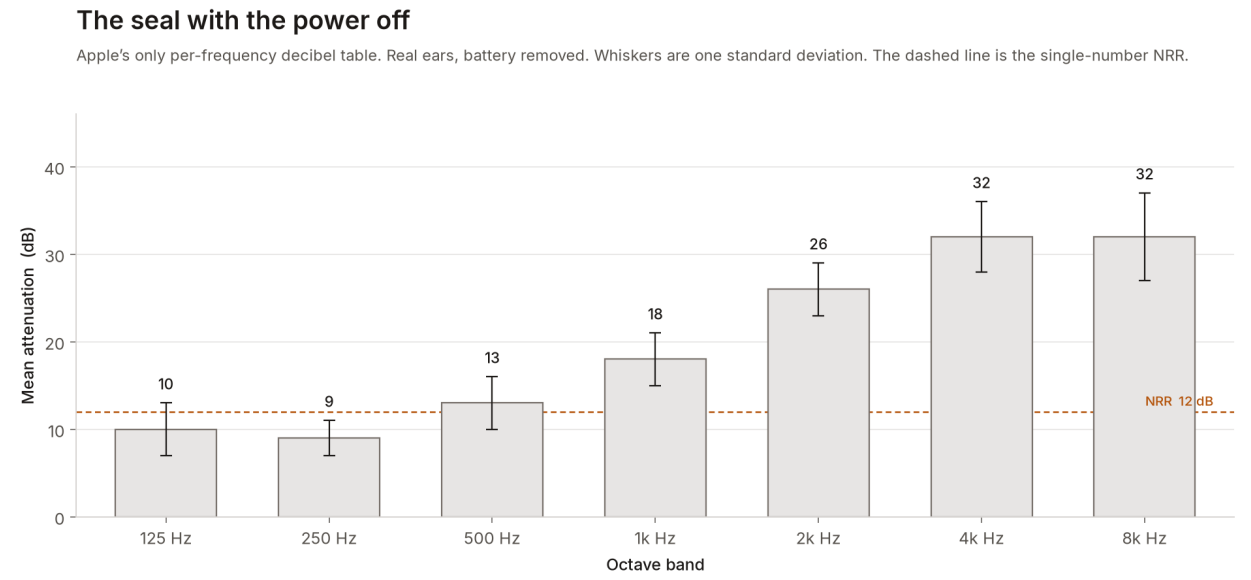


Figure 3 Apple, September 2025. ANSI S3.19-1974, unpowered. Standard deviations are 3, 2, 3, 3, 3, 4 and 5 dB. This is not the ANC curve.

With the battery out, a real-ear panel averaged about 10 dB at 125 Hz and 32 dB at 4 and 8 kHz. That shape is why ANC exists. The single-number rating of 12 dB is a summary of this uneven seal, not a promise of 12 dB at every frequency. Canada's CSA class for the passive performance is C.

Playback against a preference target

Zero is the SoundGuys target, not silence. Above the line is louder than that target. The earbud has no bass or treble control.

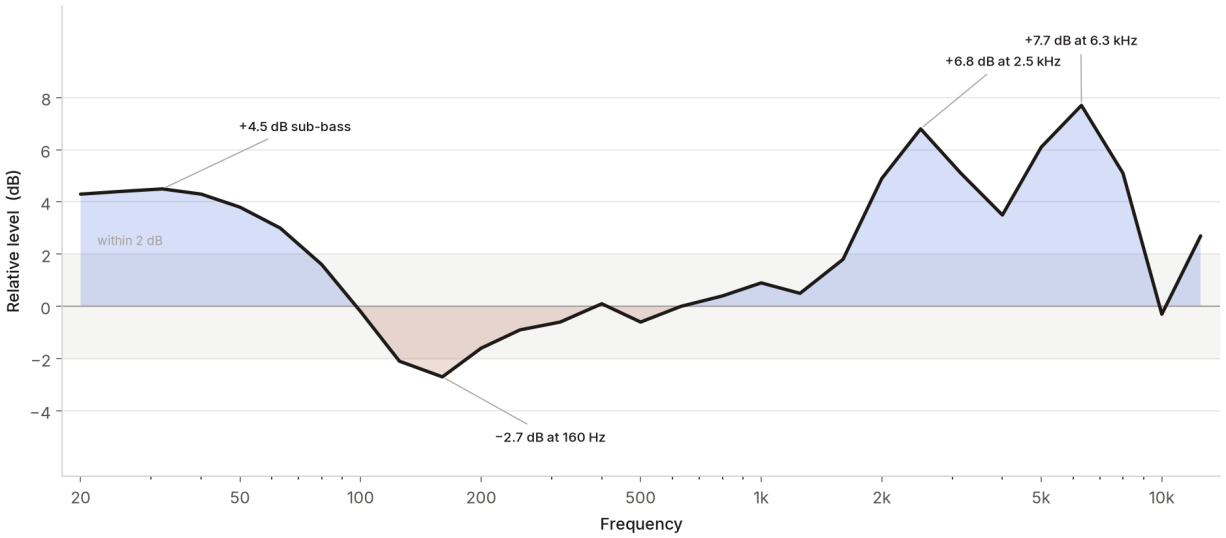


Figure 4 SoundGuys B&K 5128, relative to their preference curve. Off this scale: −11.5 dB at 16 kHz, −36.6 dB at 20 kHz.

There is no graphic equalizer, and no bass or treble slider, on AirPods Pro 3. Adaptive EQ retunes the driver from the inward microphone so a loose seal does not thin the bass. The wearer cannot edit that curve. On this laboratory comparison the sub-bass sits about 4.5 dB above the lab's target, 160 Hz sits 2.7 dB below it, and there are lifts of 6.8 dB at 2.5 kHz and 7.7 dB at 6.3 kHz. Headphone Accommodations and the Hearing Aid feature are separate, audiogram-based shapes. The iPhone Music equalizer is a phone preset. It is not stored in the earbud.

Hours on one charge

Apple's own test: 50% volume, 256 kbps AAC, preproduction units, July and August 2025. The case adds recharges, so 24 hours with the case is not drawn here.

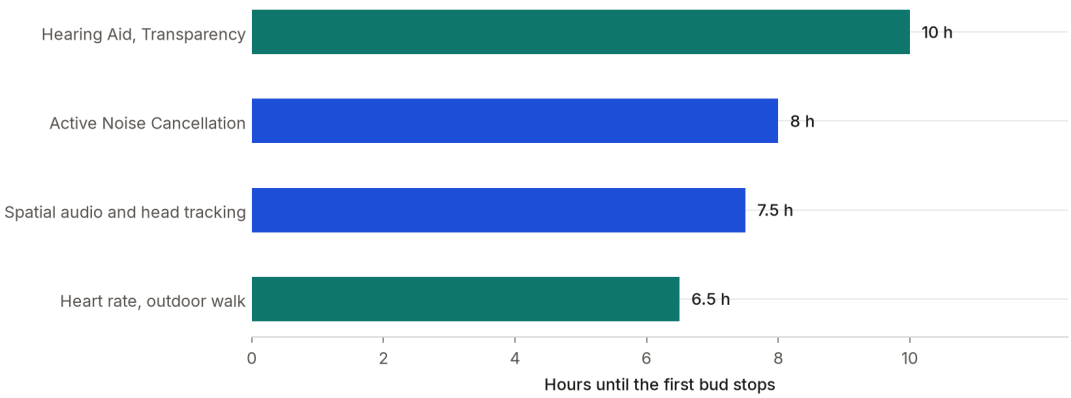


Figure 5 Apple, July–August 2025. Teal rows are Transparency. Blue rows are ANC. The 24-hour total with the case is off this scale.

Eight hours is the ANC figure at half volume. Spatial audio with head tracking is 7.5 hours. Heart-rate sensing during a walk is 6.5. Transparency with the Hearing Aid feature, and nothing playing from the phone, is 10. With the case, ANC listening is up to 24 hours, and five minutes in the case is about one hour. Putting 24 on the same axis would flatten the differences that matter on a single charge.

The stem

There is no mechanical button. Each stem is a force sensor. A press is pressure on the stem, not a click. A light swipe on the same stem changes volume. Both stems can do every gesture. Press-and-hold is the one gesture that can be pointed at Siri instead of the listening modes.

Gesture	What it does	Reassigned?
Press once	Play or pause. Answer a call. Mute or unmute. Camera remote in a compatible app on iOS 26 or iPadOS 26.	Mute and end-call can swap with the double press.
Press twice	Next track. End a call, in the default setup.	End-call can swap with mute. Skip stays.
Press three times	Previous track.	No.
Press and hold	Cycle listening modes. The default cycle is ANC and Transparency.	Per stem: modes, or Siri. The cycle can add Adaptive Audio and Off.
Swipe up or down	Volume.	Not described as reassignable.
Hold both stems	Start Live Translation.	Fixed.
“Siri” or “Hey Siri”	Playback, calls, directions, and other Siri actions.	The hold can also call Siri.
Nod, or shake	Accept or decline an announced call or notification.	Only with Announce Calls and Announce Notifications on.

Automatic Ear Detection is on by default. Remove one earbud and playback pauses. Remove both and it stops, and does not resume by itself. With neither bud in the ear, audio comes from the phone. One bud in the ear is enough for music and for a call, because each bud has its own microphones. Turning detection off keeps the audio in the buds whether or not they are worn.

Six microphone channels

Apple lists two outward beamforming microphones and one inward microphone. Read once per earbud, the pair has six channels. The outward pair is shared: it aims at the mouth for calls, and it samples outside noise for feedforward cancellation. The inward microphone is shared too: it hears what the seal failed to stop, and it hears what the driver actually delivered so Adaptive EQ can correct the bass. They are not split into “two for ANC and four for voice.”

Sensor, each earbud	Count	Job
Outward beamforming microphones	2	Voice toward the mouth. Outside noise for feedforward ANC.
Inward microphone	1	Feedback ANC, and Adaptive EQ from the sound in the canal.
Speech-detecting accelerometer	1	Own-voice vibration. Not a microphone, and not speech-to-text.
Motion-detecting accelerometer	1	Head tracking, and the nod and shake gestures.
Skin-detect sensor	1	In-ear detection.
Heart-rate sensor	1	Optical pulse sensor for workouts on iOS 26. Not part of the audio path.

Speech, and what is actually decoded

The earbuds do not turn an arbitrary song, podcast, or meeting into a transcript. Where language is recognized, the paired iPhone does that work. When the earbuds speak, the words were written on the phone and played back.

Feature	What comes back	Where it runs
Siri	A spoken answer in the earbuds.	Wake can be recognized with the H2. The request is handled on the Apple device.
Announce Calls	The phone reads the caller aloud. A nod answers.	Text-to-speech on the phone.
Live Translation	Translated speech in the ear, and a live transcript of your words on the iPhone.	Apple Intelligence on a nearby iPhone, iOS 26 or later. Hold both stems. Beta.
Conversation Awareness	No text. Media drops and the mode opens when you start speaking.	On the earbuds, from the microphones and the speech accelerometer.

Feature	What comes back	Where it runs
Hearing Aid	Louder speech from your audiogram. No transcript.	On the earbuds. FDA-authorized, age 18 or older, mild to moderate loss.
Live Listen	The iPhone microphone, played in the earbuds.	The phone is the remote microphone. No transcription.

Live Translation languages on the Apple Store page retrieved 5 November 2026: Mandarin simplified and traditional, English of the UK and the US, French of France, German of Germany, Italian, Japanese, Korean, Portuguese of Brazil, and Spanish of Spain. At the September 2025 announcement the set was English, French, German, Portuguese, and Spanish.

What is not in this note

Driver diameter, impedance, sensitivity, distortion, latency in milliseconds, an official frequency-response curve, an official ANC curve against frequency, a Transparency frequency curve, microphone part numbers, and any user bass or treble slider. Apple lists studio-quality recording and does not publish the sample rate or the bit depth. A later firmware can move a laboratory curve. The official product photograph stays on apple.com/airpods-pro. This note does not copy it.

Sources

Apple technical specifications, apple.com/airpods-pro/specs and support article 125135. Apple Newsroom, 9 September 2025. Apple Store language list, retrieved 5 November 2026. Apple Support, controls and gestures, and settings article 108764. Apple Hearing Protection data sheet for AirPods Pro 3, September 2025, regulatory file A3063 / A3064, and support article 120850. SoundGuys, AirPods Pro 3rd Gen laboratory tables, retrieved 5 November 2026.